

Bridget Liesman

Introduction

“Taylor Swift is the girl-next-door/millionaire, the hopeless romantic/savvy business-woman, and a silly teenager/serious adult singer-songwriter” (Fogarty and Arnold, 2021). This is the year of Taylor Swift, or maybe that was last year, or there is a good chance it will be next year. Taylor Swift is a central and revered figure of pop culture, and she continues to rise. She has won more Grammy Awards for Album of the Year than anyone in history. She churns out music, not even letting a global pandemic stop her from releasing two albums in the same year. Even when Swift re-releases songs that have been available for over a decade, they top the Billboard 100. Though her lyrical abilities certainly play a part in her meteoric popularity, Swift’s shrewd marketing strategy of brand reinvention must contribute to her expansive fame. She has been a darling of the Country music industry, a bubblegum pop sensation, a soulful folk artist, a victim and a villain, and a world-wide entertaining machine. What do Country’s Tim McGraw, Rock’s Panic! At the Disco, the indie-folk band Bon Iver, and rapper Kendrick Lamar all have in common? Taylor Swift has collaborated with each one. Yet, with such diversity of genre and sound, Taylor Swift’s original fans remain loyal, even as that base grows. How has Swift managed to draw new fans, while keeping the old? Despite the frequent reinventions, is there actually a common beat to her music?

Various cultural researchers and linguistic analysts have explored Swift’s wider “brand”, noting the loss of her country-twang, her softening of the girlish curls that stylized her debut, or the subtle shifts in live performances. Fogarty and Arnold (2021) explored how Swift’s wider persona models American exceptionalism, for the good and bad. While Lyon (2019) analyzed Swift’s linguistic choices in interviews across her career, Sloan (2021) tackled Swift’s

songwriting work, stating that exploring her first genre transition will, “show how Swift’s ‘calculated’ move from country to pop was as much an act of artistic experimentation and an exercise in songwriting craft as it was a savvy rebranding.” Setting aside both the intangible aspects of Swift’s brand and the specifics of her word-choice, the objective of this paper is to focus on the musical characteristics of her songs. Through analysis of music-centric aspects of Swift’s work such as acousticalness, energy, and danceability, this paper aims to determine her sound’s common themes across tracks and albums.

Data Source

To accomplish these goals, this project explored the data from Kaggle user Nicolas Fierro, who utilized the Spotify API to pull song characteristics from the app. The dataset includes song characteristic values for songs released between December 31, 1985 and August 30, 2023. These dates cover the entirety of Taylor Swift’s discography, which began on October 24, 2006 and includes her most recent album, *Speak Now (Taylor’s Version)*, which was released on July 7, 2023. As this data was pulled from Spotify, there are a few Taylor Swift tracks, such as “Sweeter Than Fiction (Taylor’s Version),” that have exclusive releases with non-Spotify retailers, such as Target, and therefore are omitted.

This dataset includes both track and album information, for example release dates, album titles, collaborators, and song characteristic rankings. These rankings are Spotify-generated quantified scores for danceability, energy, loudness, speechiness, acousticalness, instrumentalness, liveness, tempo, and valence.

Data Cleaning

A binary “taylor” variable was created that returned success (1) if Taylor Swift was the principal artist on a track and failure (0) if Taylor Swift was not the principal artist. As the

purpose of this project was to explore Taylor Swift's original music, tracks on which she was featured, but not the principal artist were coded as failure.

This project will use the song characteristic rankings for predictor variables. Five songs that had missing values for these characteristics were removed. None of the five removed tracks were by Taylor Swift. After the removal of the five tracks, the dataset held 11,445 entries, 198 of which were Taylor Swift tracks. In order to avoid the imbalanced number of failures and successes, 198 random failure tracks were selected from the dataset. This resulted in a subset of 396 tracks split equally among success and failure. Finally, this subset was again randomly split into training and testing subsets. The training subset held 80% of data (317 tracks), while the testing subset held 20% of data (79 tracks).

Results

I created a logistic regression model from the training dataset to predict if Taylor Swift is the principal artist on a song using the predictor variables acousticness, instrumentalness, danceability, energy, liveness, loudness, speechiness, tempo, and valence. The response variable y is: $y = \begin{cases} 0: \text{Taylor Swift is not principal artist} \\ 1: \text{Taylor Swift is principal artist} \end{cases}$. To determine if the logistic regression model adequately fit the data, I performed the Hosmer-Lemeshow test on the model and calculated a p-value of 0.1086. This allowed me to conclude that there was not enough evidence to show the logistic regression model does not adequately fit the data and so I will continue with this method.

After confirming that the model fit the data, I applied the AIC backwards selection method for variable selection, and determined that danceability, acousticness, and energy should be removed from the model. Utilizing this information, I built a reduced logistic regression model and compared it to the full model to determine if any of the slopes of the three removed

variables were non-zero. This test produced a p-value of 0.4911 and therefore, I concluded that there is not enough evidence to believe any of the removed slope coefficients are non-zero and so I will use the reduced model. Moving forward with the reduced model, I computed VIF values for the remaining predictor variables to check for multicollinearity. All VIF values were significantly less than ten and therefore there is no multicollinearity affecting the predictor variables of this model. This variable selection process resulted in the following logistic regression model,

$$\ln\left(\frac{\hat{\pi}(x)}{\hat{\pi}(x)-1}\right) = 3.8932 - 17.4060x_{Instrument} - 4.4608x_{Live} + 0.1438x_{Loud} - 10.2434x_{Speech} + 0.0072x_{Tempo} - 4.4621x_{Valence}$$

As this model included a binary response variable, I built a ROC Curve to determine the optimal cutoff for $\hat{\pi}_0$ (Figure 1). This curve has an AUC of 0.8276.

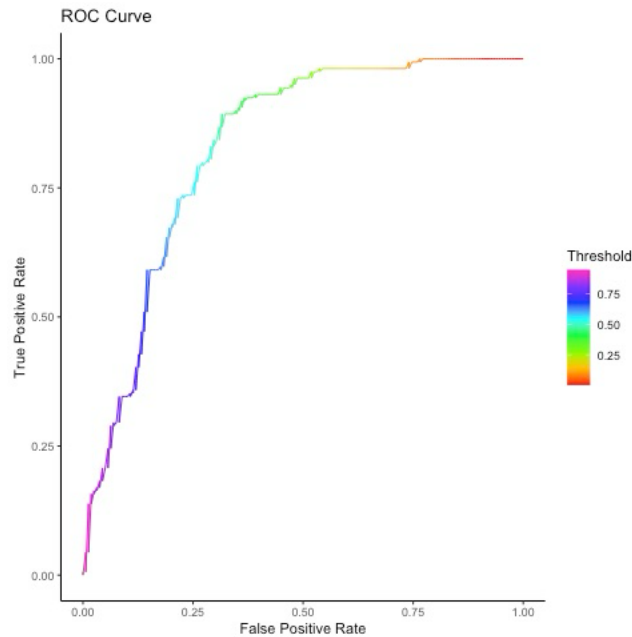


Figure 1

After reviewing the ROC curve, I chose an optimal cutoff of 0.58 for $\hat{\pi}_0$. As this data does not have an obvious reason for controlling either the true positive rate or false positive rate, the

cutoff is the value at which the TPR and FPR are equal. Finally, I utilized the testing dataset to determine the accuracy of this model. The logistic regression model, with a cutoff of $\hat{\pi}_0 = 0.58$, accurately classified 49 of the 76 datapoints, an accuracy rate of 64.47%.

Conclusion

After completing the analysis, it appears that Taylor Swift's music can be differentiated using instrumentalness, liveness, loudness, speechness, tempo and valence. However, an AUC of 0.8276 and 64.47% prediction accuracy rate suggests that the model has shortcomings in truly representing Swift's music. Due to this, there may in fact not be a definitive way to characterize Swift's sound. As these values are based on randomly selected data from a larger dataset, the process could be repeated with other random selections to more thoroughly determine the accuracy of the model. Alternatively, this model would change if the entire dataset had been used. Other metrics, outside the scope of the course, would be helpful in analyzing a model from the larger dataset.

As Swift releases more music, the increase in data points may be helpful to pinpointing certain characteristics. Additionally, it may not be possible to exemplify her entire discography through a logistic regression model, however there may be interesting time series implications to track when in time the variables of her music changed. Additionally, future analysts may more deeply explore each album to determine if intra-album tracks have greater similarities in sound. Overall, Swift's popularity may stem from her ability to create music that does not adhere to a simple definitive sound, but rather jumps between musical indicators to appeal to wider groups.

Citation

Fierro, Nicholas. (2023, August). Spotify Complete 1986 to 2023. Retrieved 28 March 2024

from <https://www.kaggle.com/datasets/nicolasfierro/spotify-1986-2023/code>

Fogarty, Mary & Arnold, Gina. (2021). “Are You Ready for It? Re-Evaluating Taylor Swift.”

Contemporary Music Review, 40(1), 1-10.

<https://doi.org/10.1080/07494467.2021.1976586>

Lyon, Lela R. (2019). “Shifting Personas: A Case Study of Taylor Swift.” *Theses and*

Dissertations--Linguistics. 33. https://uknowledge.uky.edu/ltt_etds/33

Sloan, Nate. (2021). “Taylor Swift and the Work of Songwriting.” *Contemporary Music*

Review, 40(1), 11–26. <https://doi.org/10.1080/07494467.2021.1945226>

Thompson W (2023). *taylor: Lyrics and Song Data for Taylor Swift's Discography*. R package

version 3.0.0, <https://github.com/wjakethompson/taylor>,

<https://taylor.wjakethompson.com>